

AWS Glue vs EMR vs EMR Serverless

Serverless: A Comparison



AWS Glue, Amazon EMR (Elastic MapReduce), and EMR Serverless are all services offered by Amazon Web Services (AWS) for data processing and analytics, but they serve slightly different purposes and have different characteristics. Let us have a detailed look at this data analytics pipeline showdown.

Overview

AWS Glue is a fully managed extract, transform, and load (ETL) service provided by Amazon Web Services (AWS). It simplifies and accelerates the process of preparing and transforming data for analytics by automating many of the time-consuming tasks involved. With features like automated schema discovery, data cataloging, and job orchestration, AWS Glue enables users to efficiently integrate, clean, and transform data from various sources, making it ready for

analysis. This service is particularly beneficial for data engineers, analysts, and data scientists who require a seamless solution to manage data pipelines and ensure data quality before further processing.

Amazon EMR (Elastic MapReduce) is a cloud-native big data platform provided by AWS and is one of the oldest services provided by AWS. It enables businesses to process and analyze vast amounts of data efficiently by leveraging distributed computing frameworks like Apache Hadoop, Spark, and Hive. EMR offers fully managed clusters that automatically scale based on workload demands, allowing organizations to tackle complex data processing tasks without the need to manage infrastructure. The service supports a range of use cases, including batch processing, data warehousing, log analysis, machine learning, and more.

AWS EMR Serverless is a relatively new offering within Amazon EMR (Elastic MapReduce) that focuses on delivering serverless data processing capabilities. It allows users to effortlessly run big data analytics and queries without the need to manage clusters or provision resources. With EMR Serverless, resources are automatically allocated based on workload demands, optimizing cost and efficiency. This approach is ideal for ad-hoc querying and interactive analytics, enabling users to quickly access and analyze data without the complexities of cluster setup. By combining the power of EMR's distributed processing capabilities with the convenience of serverless computing, AWS EMR Serverless offers a flexible and cost-effective way to process data, making it well-suited

for organizations seeking on-demand data processing without the burden of resource management.

Features Comparison

Given below is a table that I have compiled from my research, experience and understanding of AWS data analytics stacks. Please note that the views are mine, from the extend of exposure I have had on these stacks.

Feature	AWS Glue	Amazon EMR	AWS EMR Serverless
Initial Release	2017	2009	2021
Purpose	Data integration, ETL, and data preparation, with a Data Cataloging feature.	Processing and analyzing large-scale data.	A deployment option for EMR. Serverless data processing for ad-hoc queries.
Managed Service	Yes, fully managed by AWS.	Yes, fully managed by AWS.	Yes, fully managed by AWS.
Serverless	Yes, Automatically manages resources based on workloads.	No, Requires manual or automated cluster setup. Can be configured to run on EC2s or EKS.	Yes, Automatically manages resources based on workloads.
Scalability	Automatically scales resources for transformations.	Scalable clusters for distributed processing, leveraging frameworks similar to EC2/ EKS autoscaling.	On-demand resource allocation for processing.
Cost Efficiency	Pay-per-use. Cost-efficient due to resource allocation.	Provisioned to right-size. Offers cost savings with spot instances.	Pay-per-use. Cost-efficient due to resource allocation.
Data Processing	Handles ETL and data transformation.	Processes data using distributed frameworks.	Suitable for ad-hoc queries and analytics.
Big Data	Can handle large datasets for transformations.	Specifically designed for big data processing.	Handles large datasets for processing.
Frameworks Supported	Only Spark and Py Shell	Many, including Spark, Hadoop, Hive, Pig, Presto, Sqoop, HBase etc.	Many, including Spark, Hadoop, Hive, Pig, Presto, Sqoop, HBase etc.
Scripting	Supports Python or Ray (for Spark)	Supports various scripting languages like Java, HiveQL, Pig, Cascading, Ruby, Perl, Python, R, PHP, C++, and Node.js	Supports various scripting languages like Java, HiveQL, Pig, Cascading, Ruby, Perl, Python, R, PHP, C++, and Node.js
Ease of Use	Simplified with automated ETL workflows.	More management required for custom clusters.	Abstracts cluster setup and management.
Customization	Limited customization due to serverless nature.	Extensive customization and tuning options, including bootstrap installations.	Limited customization due to serverless nature.
Local Persistent Storage	None. Other services like S3, Dynamo, Redshift etc. can be used.	Yes, Besides other AWS storage integrations HDFS and local file system storages are provided.	No HDFS support, and only temporal storage on local. Other services like S3, Dynamo, Redshift etc. can be used.
Metadata store	Glue Data Catalog and Crawlers are supporting features.	Leverage Glue Data Catalog or an external Hive metastore (like RDS)	Leverage Glue Data Catalog or an external Hive metastore (like RDS)
Interactive Workloads	Yes, through Glue Studio, Sagemaker Studio and notebook services.	Yes, through EMR Studio or other notebook services.	No interactive support yet.
Complexity	Provides simplicity for ETL tasks.	More complex due to distributed computing.	Simplifies resource management and scaling.

Conclusion

In summary, AWS Glue is best for being fully managed and serverless for ETL tasks, data preparation, and building data pipelines.

Amazon EMR is suitable for processing and analyzing large datasets using distributed frameworks especially beyond Spark, like Hadoop, Hive, Presto etc.

EMR Serverless focuses on the serverless aspect of EMR — data processing, offering on-demand scaling and cost optimization.

The choice between these services depends on your specific use case and requirements.